

Gastric Cancer Histopathology Classification Project

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1) Project Overview

This project implements a medical image classification framework for multi-class gastric cancer histopathology. It uses PyTorch with a ConvNeXt ensemble, Grad-CAM interpretability, and comprehensive training and evaluation tooling. The target classes are ADI, DEB, LYM, and MUC.

2) Model Architecture

The core model is a ConvNeXt ensemble composed of ConvNeXt-Tiny, ConvNeXt-Small, and ConvNeXt-Base backbones. Each backbone replaces the default classification head with a dropout layer and a linear output layer matching the four classes. The ensemble output is computed by stacking each model output and applying learnable weights for weighted averaging.

3) Model Factory & ViT alternatives

The project exposes a model factory function to build either a single ConvNeXt classifier or the full ensemble. Supported model names include `convnext_tiny`, `convnext_small`, `convnext_base`, and `convnext_ensemble`. A Vision Transformer (ViT) alternative is not currently implemented in this repository, but a natural future extension would be to add ViT backbone support via `torchvision` or `timm`, allowing direct comparison between ConvNeXt and transformer architectures.

4) Training Configuration

Training is configured in `config.yaml`. Key settings include:

- `num_classes`: 4
- `image_size`: 224
- `batch_size`: 32
- `model ensemble`: `convnext_tiny`, `convnext_small`, `convnext_base`
- `optimizer`: AdamW, `lr`=0.0001, `weight_decay`=0.0001
- `scheduler`: CosineAnnealingLR
- `dropout`: 0.2
- `epochs`: 20
- `early stopping patience`: 8

5) Evaluation Results

Evaluation generates detailed metrics including accuracy, precision, recall, macro F1-score, log loss, and multi-class ROC-AUC. The repository includes reports such as `test_metrics.json`, confusion matrices, per-class metrics plots, and ROC curve visualizations. Grad-CAM visualizations are also produced for model interpretability on test images.

6) Interpretation & Insights

Interpretability is provided through Grad-CAM attention maps, which highlight regions contributing most to each prediction. This allows practitioners to verify that the ensemble is focusing on relevant histopathological structures rather than background noise. The ensemble approach improves robustness by combining the strengths of lightweight, balanced, and higher-capacity ConvNeXt variants.

Overall, the project is built as a practical research pipeline with configurable training, evaluation, visualization, and inference components suitable for gastric cancer histopathology classification tasks.